

# BHUVANESHWARI B

AI Backend Engineer

Mumbai, India

bhuvanabaskaran056@gmail.com — +91-8189967315

## Professional Summary

AI Backend Engineer with hands-on experience building and deploying production-grade Generative AI systems in enterprise environments. Strong expertise in LLM integration, OCR-to-LLM pipelines, FastAPI-based backend services, and multi-database architectures. Proven track record of improving AI reliability through confidence scoring and human-in-the-loop validation. Experienced in collaborating with senior engineers, handling production issues, and presenting complex AI systems to senior leadership. Currently contributing to medical-domain AI platforms preparing for international deployment.

## Technical Skills

**Programming & Backend:** Python, FastAPI, REST APIs, RPC / Nameko, Microservices Architecture

**Generative AI & NLP:** Large Language Models (LLMs), OCR + LLM Pipelines, Tesseract OCR, Prompt Engineering, RAG (Retrieval-Augmented Generation), Human-in-the-Loop Systems, Confidence Scoring, AI Reliability, MCP

**Databases:** PostgreSQL, MongoDB

**DevOps & Enterprise Tooling:** Git, Bitbucket, Jira, Confluence, Jenkins (CI/CD Exposure), OpenShift (Deployment Exposure)

**Frontend (Support):** React.js (UI Fixes, API Integration Support)

## Professional Experience

**Software Engineer – AI Backend**

Dec 2024 – Present

Capgemini, Mumbai, India

- Backend engineer on a production medical-domain AI platform, developing and maintaining FastAPI-based services supporting OCR and LLM-driven workflows.
- Integrated Tesseract OCR outputs with LLM pipelines to clean, structure, and interpret medical text data with high reliability.
- Designed robust backend APIs with strong input validation, response normalization, error handling, and enterprise-grade stability.
- Implemented confidence-based scoring mechanisms to detect low-reliability AI outputs and route them into human-in-the-loop review flows.
- Collaborated closely with senior backend engineers (10–15+ years of experience) through architecture discussions, deep code reviews, and production issue debugging.
- Gained hands-on exposure to CI/CD pipelines using Jenkins and containerized deployments via OpenShift, understanding enterprise release workflows.
- Actively used Git, Bitbucket, Jira, and Confluence within Agile enterprise environments.
- Supported frontend teams during critical production situations, contributing React.js UI fixes and API integration support to unblock releases.
- Built and integrated microservices using Nameko and RPC-based communication for service-to-service interactions.
- Presented the GenAI system to senior leadership (C2+ level), explaining architecture, OCR–LLM workflows, AI reliability design, and business impact.
- Contributing to a system preparing for international production deployment, focusing on backend robustness, failure handling, and API stability.

## Key Projects

**Medical OCR + LLM Processing Platform**

*Python, FastAPI, Tesseract OCR, LLMs, PostgreSQL, MongoDB*

- Backend engineer for a production medical AI application combining OCR-extracted text with LLMs for structured interpretation and insights.
- Developed APIs for OCR processing, prompt preparation, LLM response parsing, and persistence across relational and NoSQL databases.
- Implemented confidence-driven validation pipelines to improve AI reliability and support manual review workflows.
- Followed enterprise standards for documentation, deployment, monitoring, and compliance.

#### **GenAI Issue Analysis Platform**

*Python, LLMs, RAG, Jira Integration*

- Built an AI backend platform for automated issue understanding and root cause analysis integrated with Jira APIs.
- Implemented conversational querying and RCA workflows using LLMs and retrieval-augmented generation.
- Added confidence scoring and human-review checkpoints to reduce hallucinations and improve enterprise trust.

#### **Multi-Agent Climate-Aware Routing System**

*AWS Bedrock, Lambda, Step Functions, DynamoDB*

- Designed a multi-agent AI system to optimize routing decisions under environmental and efficiency constraints.
- Integrated external APIs and built serverless workflows using AWS managed services.

#### **LLM-Based Code Migration System**

*Python, Prompt Engineering*

- Designed chunk-based LLM pipelines to migrate large codebases under token limits.
- Implemented overlapping chunk strategies and post-processing logic to maintain code continuity and reduce duplication.

## **Education**

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**B.E. in Biomedical Engineering**

GKM College of Engineering and Technology, Chennai

2021 – 2024